

MODULE CODE	CCS6213
MODULE TITLE	Back-end Application Development
PROGRAMME	Web and Mobile Development
DEPARTMENT	Computer Science
CREDITS	15
STAGE OF STUDY	PG
SEMESTER/SESSION	Spring 2025-26
LOCATION	Athens
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ACCREDITATION	N/A

ASSESSMENT NUMBER	1
CONTRIBUTION	60% of the module final mark
ASSESSMENT TITLE	Semester Long Project
ASSESSMENT TYPE	Individual
HAND-OUT DATE	17/March/2026
SUBMISSION DATE	21/April/2026
FEEDBACK DATE	12/May/2026

LEARNING OUTCOMES
Upon completion of this piece of coursework, a student will be able to: ▪ Build applications using Java Spring ▪ Build Restful APIs using Java Spring ▪ Understand the benefits of service-oriented design and software parallelism ▪ Familiarise with RDBMS design and PostgreSQL ▪ Take advantage of asynchronous code ▪ Apply caching techniques ▪ Apply testing techniques successfully ▪ Assess security risks and implement robust software

ASSESSMENT CRITERIA
▪ Software efficiency and API design (parallelism, speed, accuracy, etc) (40%) ▪ Security (Authentication, throttling) (20%) ▪ Test coverage (20%) ▪ Report with Code comments, API documentation, justification of design decisions and tests (20%)

DETAILED DESCRIPTION

Part 1

Taking the role of a backend engineer, you are requested to design and develop a system that will manage books.

A book entity must have a title, a publication date, a category, an author and the number of pages in it.

An author entity must have a first name, last name, country and number of books published.

Your system should fulfil all the following requirements:

- Using a PostgreSQL (docker image), design your database tables that will store your books
- Support all CRUD (Create, Read, Update, Delete) operations
- Provide a search endpoint that will allow the API caller to search based on all book attributes (title, author, etc)
- Provide the following analytics:
 1. **Category Distribution:** Calculate the average number of pages and the total number of books for each category, ordered from highest to lowest volume.
 2. **Author Productivity:** Retrieve a timeline report showing the number of books published per year for a specific author.
 3. **Geographic Reach:** Calculate the distribution of authors by country, including the total number of books originating from each country.
- Design appropriate testing functionality to test the following operations:
 1. Test the "Read" operation by successfully creating a single book and then fetching it by its unique ID to ensure the API returns a 200 OK and the exact same data that was just saved.
 2. Test the "Delete" operation by removing a specific book and verifying two things: that a subsequent request for that book returns a 404 Not Found, and that the associated Author entity was *not* accidentally deleted from the database along with it.
 3. Test the system's resilience by attempting to create a book with invalid data (e.g., a non-existent author ID or a negative page count), ensuring the API rejects the request and returns the appropriate HTTP status codes (4XX).
- All endpoints provided should respect the RESTful principles with appropriate HTTP codes for relevant errors, etc

NOTE:

Do not make a front-end app, this coursework does not concern itself with a UI.

Part 2

Extremely pleased from your performance and the previous deliverables; the company is now requesting you to extend upon the previously submitted work.

During the project briefing, the company is sharing some security concerns with you. They would like you to enhance the previous endpoints with a token-based authentication system and incoming throttling (20 Requests per 2 Seconds) in order to avoid any surprises.

Also, they are providing you with an external service that they would like to consume. This is an External REST API that provides IP address lookup and related services. Using this API (<https://ipapi.co/>), you are requested to protect the creation and deletion operations and restrict

the origin of the requests to Greece. Any request coming for create and delete from any other country should be denied.

Other than that, you are requested to enhance all relevant services and/or repositories with in-app caching to optimise the usage of the persistent layer and external dependencies. Provide and implement it for all use cases.

In order to keep up with the high quality of your previous work, you should focus on delivering fast, high-quality code within specs. Proper HTTP responses/codes and good test coverage (above 80%)

SUBMISSION

Students are expected to submit their work to private repository in [Athens Tech College](#) GitHub. Late submissions will result in a 10 marks reduction for each day that the work is late, up to a total of five days. (see student handbook for more information)

Make sure you also provide a Postman collection and instructions on how to setup and install your application.

NOTE

This piece of assessment should be completed and submitted by the student (or group of students in group work) without assistance from or communication with another person either external or fellow student (outside the group). All sentences or passages cited in the assignment from other people's work should be specifically acknowledged by complete and accurate reference to the author, work and page(s). Failure to abide by the above regulation constitutes use of unfair means (collusion, plagiarism etc.) and will result in a fail mark for this work. It might also invoke disciplinary actions. It is at the instructor's discretion to conduct an oral examination, which will result in the award of the final grade for that particular piece of assessment.